

1. Basic Linux Commands

ls : lists the files and folders inside current location. (Sample Usage: **ls -la**)

cd : moves you into a different folder. (Sample Usage: **cd /home/user/Documents**)

pwd : shows the exact folder path you are in. (Sample Usage: **pwd**)

mkdir : creates a new folder. (Sample Usage: **mkdir new_project**)

rm : deletes files or folders from your computer. (Sample Usage: **rm -rf old_folder**)

cp : copies files or folders from one place to another. (Sample Usage: **cp config.txt backup_config.txt**)

mv : moves files to a new location or renames them. (Sample Usage: **mv old_name.txt new_name.txt**)

cat : reads a file and shows its text on the screen. (Sample Usage: **cat error_log.txt**)

grep : searches for a specific word or phrase inside a file. (Sample Usage: **grep "failed" server.log**)

chmod : changes the rules for who can read, edit, or run a file. (Sample Usage: **chmod 755 script.sh**)

2. Basic Git Commands

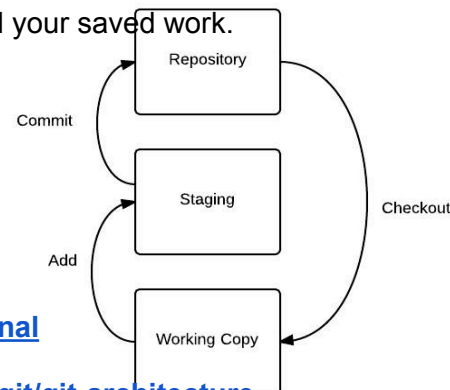
Command	Sample Usage	Description	Options
git init	git init	Starts a new empty git project	-bare
git status	git status	Shows which files you changed	-s
git add	git add .	Puts changed files into a waiting area so they are ready to be saved	-p
git commit	git commit -m "fix"	Permanently saves your changes	-a, --amend
git log	git log	Shows list of past saves made	--online
git diff	git diff	Shows exact lines of text changed	--staged
git restore	git restore f.txt	Undoes recent change or takes a file out of waiting area	--staged
git branch	git branch feat	Lists branches or makes a new branch	-d, -a
git switch	git switch main	Lets you switch different branches	-c
git merge	git merge feat	Combines work from branch to another branch	--no-ff
git tag	git tag v1.0	Gives a name to a specific save point	-a
git clone	git clone url	Downloads a full working copy of project from net to pc	--depth 1

3.The three-tree architecture of Git

Working Directory: these are the actual files you see and edit on your computer right now.

Staging Area: this is like a waiting room. You place the specific files here that you want to include in your next save.

Repository: this is the hidden database where git keeps a permanent record of all your saved work.



4. References

<https://git-scm.com/tools>

<https://ubuntu.com/tutorials/command-line-for-beginners#3-opening-a-terminal>

<https://www.darwinbiler.com/understanding-the-3-tree-architecture-of-git/>

<https://nulab.com/learn/software-development/git-tutorial/git-basics/what-is-git/git-architecture/>

I have used Gemini to better understand the description of each command. I edited the content and used my own words to give the description of the commands

